Mini-Project Milestone 2

Learning Models for Pressure Forecasting

DS675
Andres Arango Martinez
NJIT ID: 31748996 | UCID: aa3978

1. Description of Dataset and Learning Task

1.1 Source, records and suitability

BattLeDIM is an open, simulated drinking-water benchmark representing L-Town, inspired by Cyprus. It was created to compare leak detection/localization methods. Its continuous time series also allow a supervised pressure-forecasting task. Zenodo supplies supervisory control and data acquisition (SCADA) records, leak labels, sensor configuration and EPANET hydraulic models [1,2]. These are simulated observations, not field measurements.

The 2018-2019 release has five-minute spacing: 105,120 timestamps per year, 210,240 total, and 119 numerical measurement channels per timestamp. The nominal network contains 782 demand junctions, two reservoirs, one tank, 905 pipes, one pump and three pressure-reducing valves (PRVs). Nominal and generator parameters differ [1].

Quantity	Channels / units	Meaning
Pressure	33; m water head	Pressure at measured junctions.
Pipe/pump flow	3; m^3/h	PUMP_1 and p227/p235 flow.
Tank level	1; m	Water level in T1.
AMR demand	82; L/h	Automated meter reading (AMR): consumption assigned to junctions, not pipe inflow.

Our checks of all eight annual SCADA tables found matching, unique timestamps and no missing values. The benchmark assumes error-free sensors and no transmission delay/loss. Values are rounded to two decimals; pressure readings average the preceding five minutes [2], Sensors and Telemetry. Nearby time records are related, so the later evaluation keeps time order. Leak labels are not model inputs.

1.2 Learning task and hydraulic feature selection

The task is single-output supervised regression: y(u)=p_n1(u+15 min), three samples ahead, in metres. The label is the recorded five-minute average ending at u+15, including the simulated operating/leak conditions. Each input row is paired with n1 three timestamps later; there are no target classes. The intended use is a short look-ahead for operators reviewing pressure changes at a remote junction; operational benefits are not evaluated here. Every input must be available at forecast time u.

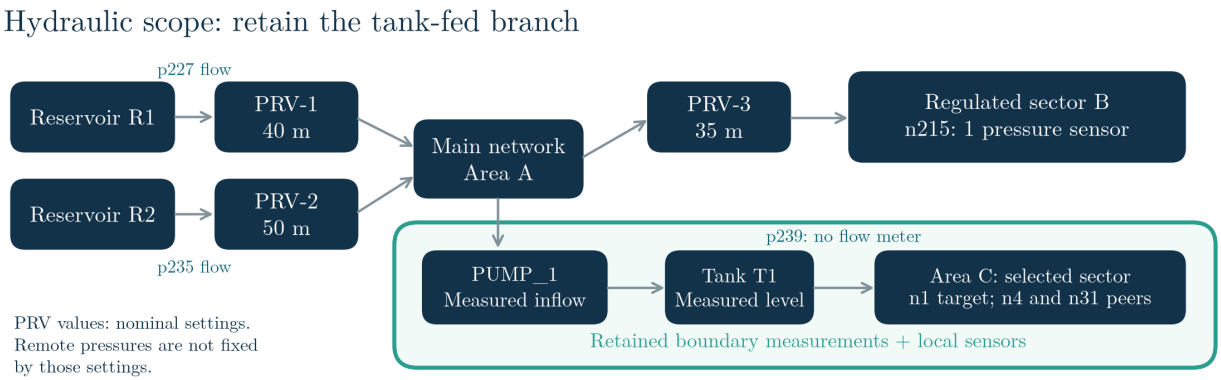


Figure 1. Functional supply boundary derived from the released model [1]. Nominal PRV settings are 40/50/35 m; constant settings do not fix remote-node pressures.

Area C is supplied through T1 and has 92 junctions, three pressure sensors and 82 AMRs. Our nominal-model checks give n1 the longest T1-to-sensor pipe path: 891 m, versus 574/308 m for n4/n31; this distance is not a signal delay. We keep n1/n4/n31 pressure, T1 level, PUMP_1 flow and AMR_n1 demand: 119 channels → 87 local/boundary candidates → six signals. T1 represents storage, PUMP_1 measured inflow, n4/n31 the local pressure state, and AMR_n1 consumption at the target junction. Unlike PRV-3's single measured pressure, C provides several related measurements. Whether they help n1 forecasting is a question to test.

1.3 Own exploratory analysis

Our exploratory data analysis (EDA) uses January-September 2018 only: 273 days and 78,624 timestamps. n1 ranges from 25.67 to 29.36 m. The generator opens PUMP_1 below T1=2.4 m and closes it above 3.9 m [1]. Figure 2 shows how pressure, tank level and pump operation vary through the day; the mean curve does not describe every individual day.

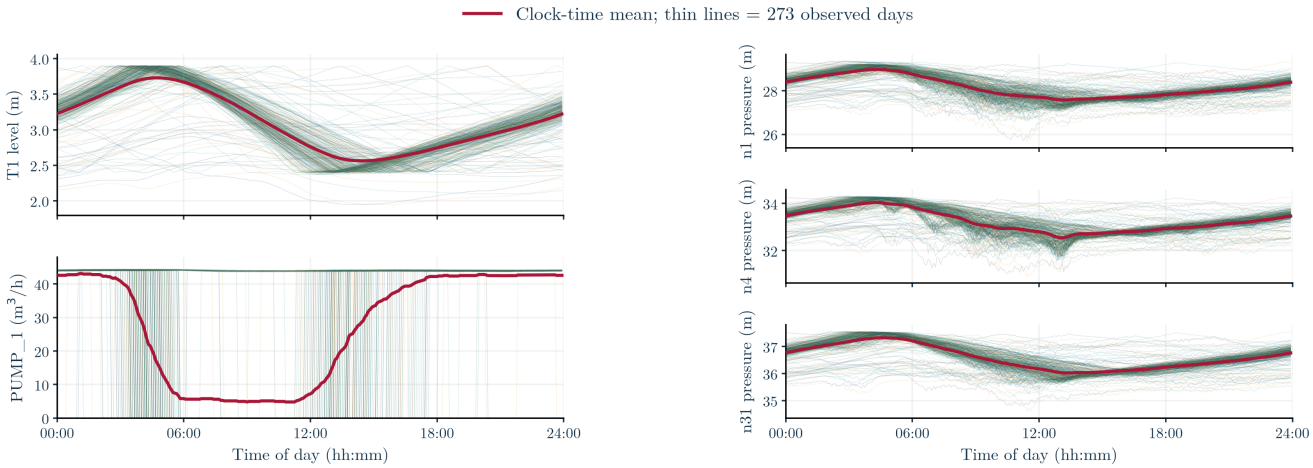


Figure 2. Own EDA: T1/PUMP_1 at left; n1/n4/n31 at right. Thin lines: individual days; red: clock-time mean.

For 78,621 valid +15 pairs, excluding the final three origins, persistence predicts the current n1 value. Its mean absolute error (MAE) is 0.0518 m (about 5.18 cm); root mean squared error (RMSE) is 0.0685 m and gives more weight to large errors. These are our 2018 descriptive results, not final test scores. Correlation with future n1 is 0.9923 for current n1, 0.9230 for T1, -0.0367 for PUMP_1 and -0.1574 for AMR_n1. Pressure changes slowly over 15 minutes, making persistence a strong baseline; high correlation alone does not show that an extra signal improves a forecast.

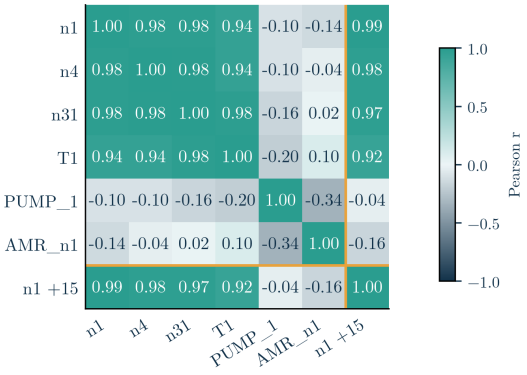


Figure 3. Own EDA: six origin inputs and the future label. The label is separated for analysis and never enters X.

2. Contributions of Previous Authors

We searched BattLeDIM/L-Town predictive models and followed citations/code. Related papers/repositories form 35 groups of related works on L-Town or explicit modifications; external networks, reviews and tools were excluded. The five categories mean leak detection/location, current-pressure estimation, future-pressure/head forecasting, anomaly/sensor-fault detection, and sensor placement. This inventory is not exhaustive.

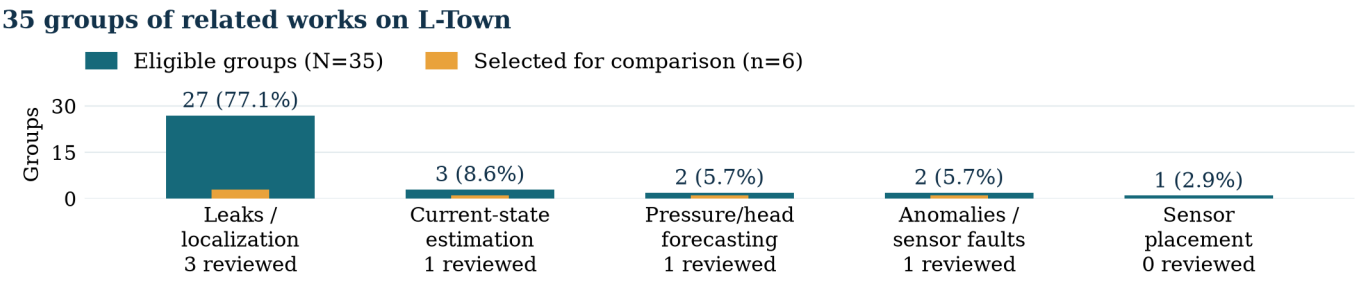


Figure 4. Own screening: percentages use 35 groups; gold identifies six selected subsets. Primary diagnostic purpose can include an intermediate future-pressure prediction.

ChebNet, SAM-kNN and AGN are core future-state studies; LILA/GATRes are current-state comparators; EILF is supporting hydraulic engineering. SAM-kNN/AGN/GATRes generate new L-Town data rather than using the released SCADA. Training was not reproduced.

Study / task / code	Data preparation and how the method works	Reported results and what they show
ChebNet [3] +5-min pressure and leakage diagnosis. Code: [4].	**Data:** Leak-free nominal simulations, network links and three past pressure readings; 80/20 train/validation split. **Models:** Two four-layer graph networks: one forecasts healthy pressure (+5 min), the other reconstructs current pressure from available sensors. Their discrepancy over seven days helps locate leaks. Graph/scaling options were compared. Code fits scaling before the split.	**2019 diagnosis, Table 1:** 19 accepted matches, 138 false alarms, four missed leaks; recall 82.61%, EUR326,521. A match requires a location within 300 m while the leak is active. Many false alarms remain; these results do not measure forecast-pressure error.
SAM-kNN [5] +5-min virtual pressure; anomalies. Code: [6]; scenarios not shared.	**Data:** Ten new Area A simulations (3 months each; 29 sensors). Code averages three past readings from other sensors to predict the target's next reading (+5 min); paper describes averaging four. **Model:** Adaptive nearest-neighbor regression stores recent/older examples and selects the memory with the lowest tracked prediction error.	**Five leak cases, Table 1:** Each leak is detected, with 3-48 false alarms per scenario versus 11,146-19,216 for ordinary online kNN. These are alarm counts, not pressure errors. The scenarios are newly simulated, not the released 2018-2019 series.
AGN [7] Multistep future nodal heads. Code: no repository link found in the inspected paper version; data on request.	**Data:** One generated year on modified L-Town: 5-min samples, 104 demand patterns; 70/20/10 train/validation/test split. Elevation and water-use analysis guides features. **Model:** Graph links share information; recurrent units retain history; attention weights nodes/features to forecast head (pressure head + elevation). Training progressively hides fewer inputs.	**Future heads, Table 1:** 12/24/48/96-step outputs reach +1/+2/+4/+8 h. For the 12-step window, regional MAE is 0.859 m (high), 0.327 m (medium), 32.7 m (low elevation). Errors average whole windows/regions; +15 is included, but no separate n1(+15) score is reported.
LILA [8] Current healthy-pressure reference; leaks. Code: [9].	**Data:** Released SCADA; pressure/flow analysis guides reference-sensor selection. Leak-free calibration periods and some per-leak node combinations are manually selected. **Model:** Linear regression estimates current healthy pressure from peer pressures/flow. CUSUM adds persistent deviations to detect leaks, then hydraulic calculations locate them. Paper: squared flow; code: raw PUMP_1. It is not a future forecast.	**Journal Table 1:** 19 new 2019 leaks; four inherited leaks excluded. TP=17, FP=2, FN=0; precision 89.5%, F1=0.944, EUR307,852. Two locations outside 300 m count as false positives, not missed events; recall=1 therefore does not mean perfect localization.
GATRes [10] Current sparse-to-full pressure. Code: [11].	**Data:** 10,000 simulated L-Town snapshots (60/20/20 train/validation/test split); compare pressure distributions with time-series test data. **Model:** Hide selected readings and fill hidden inputs with zero. Graph attention weights neighboring information to reconstruct the hidden pressures at the same time point.	**Table 10 (text: 1 week; caption: 24 h):** Across ten training runs, MAE=0.9501±0.0086 m versus 1.5928±0.0050 m for the mGCN graph baseline. The ± is the standard deviation between runs. This measures current-pressure reconstruction, not +15-min forecasting.
EILF [12] Engineering diagnosis, not future learning. Code: on request.	**Data:** 2018 development/2019 test. In Area C, pump inflow minus the tank-storage change rate estimates sector outflow; summed AMRs provide a reference. Use 30-min windows to reduce level-rounding effects. **Method:** Separate pressure trends from recurring cycles (STL), detect sustained changes, then locate leaks hydraulically.	**Journal conclusion, p. 11:** TPR=47.83%, five false positives, EUR167,981. These measure leak diagnosis, not future-pressure accuracy. The tank-balance calculation is an existing engineering contribution relevant to our feature selection.

TP: correctly identified leaks; **FP:** false alarms/incorrect matches; **FN:** missed leaks. **TPR:** recall; **F1:** balance between precision and recall. **EUR:** competition monetary score [2].

Research gap. Among the reviewed studies, none reports a directly comparable n1(+15) forecast evaluation on the released SCADA. Leak diagnosis, current reconstruction and forecasts with other data/horizons answer different questions. We therefore compare Gradient Boosting, persistence, Ridge and Random Forest on common chronological splits and test auxiliary-signal value. This provides reproducible classical baselines for future comparison with complex models, which are not reproduced here.

3. Your Anticipated Contributions

3.1 Proposed contribution

We propose to forecast recorded n1 pressure 15 minutes ahead using Gradient Boosting and the six signals selected in Section 1: n1/n4/n31 pressures, T1 level, PUMP_1 flow and AMR_n1 demand. The contribution would be a reproducible comparison for this specific task using the released SCADA data.

The reviewed graph and adaptive-kNN studies address related prediction tasks. Boosting already classifies L-Town leakage regions [13,14], and Artesia estimates night pressure with ExtraTrees without specifying a future lead [15]. Our review did not identify this exact six-signal n1(+15) experiment. We therefore propose an extension of the reviewed applications, without claiming a new algorithm or worldwide priority.

Planned contribution	Question to answer
C1. Prepared forecast data	Can we organize past measurements and their +15-minute pressure labels without using future inputs?
C2. Model comparison	Does Gradient Boosting improve on persistence, Ridge and Random Forest?
C3. Value of extra signals	Do the tank, pump, peer pressures and local demand help beyond n1 history?

3.2 Data preparation and models

We will align timestamps, check units and missing values, and pair each input row with n1 three readings later. Starting inputs will include current readings and the previous 55 minutes at five-minute intervals, plus time of day and day of week. Only information available at forecast time will be used; leak labels will stay outside the model inputs. Rows without required values or with a target outside their assigned period will be excluded. Any scaling for Ridge will be fitted on training data only. Section 1 EDA motivates these choices but does not establish that every signal helps.

Gradient Boosting adds small decision trees that try to correct earlier prediction errors [18]. We will compare it with persistence (use the current n1 pressure), Ridge (a linear model with a penalty on large coefficients) [16], and Random Forest (average predictions from many trees) [17]. These models build on regression and ensemble ideas studied in class. Model settings will be chosen using validation data; detailed parameter choices are left for implementation.

3.3 Evaluation and expected improvement

We will train on January-September 2018, use October-December 2018 to select settings, then refit on eligible 2018 rows and assess 2019. This preserves time order. Since 2019 has already been inspected, it is a final chronological assessment. All models will predict the same rows.

We will report MAE (average absolute error) and RMSE (which gives more weight to large errors), both in metres. A boosting advantage would require lower MAE than persistence, Ridge and Random Forest. To test the extra signals, we will also compare full boosting with boosting using n1 history and the same calendar variables. Extra-signal value requires lower MAE than this reduced model and persistence. We will report RMSE tradeoffs and cases where no improvement is found.

The expected outputs are an annotated preparation/model notebook, a comparison table and forecast/error plots. These are planned experiments; no learning model has been fitted.

4. References

Numbered citations link to the entries below; entries open the original article, author repository or official documentation. Screening counts and EDA are our analyses. The 2018 persistence result is a descriptive reference; the proposed models have not been trained or evaluated.

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[14] Liu, R., Zhang, Z., and Zhang, D. (DHI China, 2020). Leakage Detection and Isolation in Water Distribution Network Based on Data Mining and Genetic Optimized Hydraulic Simulation [Report]. Zenodo. doi:10.5281/zenodo.3911523.

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